Quiz 1: CS 215

INSTRUCTIONS:

1. Attempt all five questions, each carrying 10 points. Clearly mark out rough work, mention question numbers.
2. Write all answers neatly with black or blue ink. Take clear pictures of all your work in the correct order and submit a zip file or a PDF (preferable, as far as possible) on moodle BEFORE the cutoff deadline of **6:00 pm today**. Name the zip file as YourRollNumber.zip or the PDF as YourRollNumber.pdf.
3. Write your name and roll number on the top of the first page.
4. You have 1.5 hours for the quiz and up to 1 hour more for scanning and uploading your work on moodle. No submissions will be accepted after 6 pm.
5. You are allowed to refer to all class notes, slides, handouts and the class textbook as well as previous years quizzes/midsems that I shared with you. You are not allowed to browse the web or talk to each other. **By submitting this quiz, you promise that you abided by these policies.**
6. You may use any results or theorems discussed in class directly, by just stating them without proving/deriving them.
7. When in doubt, make suitable assumptions and state them clearly.

QUESTIONS:

1. A certain gene occurs in one of two forms F_1 or F_2 with probabilities p and $1-p$ respectively. Each cell of a person contains a pair of these genes, and the exact same pair is found in all cells of a given person. Out of n people tested, it was seen that the cells of n_1 people contained genes of type F_1, F_1. Cells of n_2 people contained genes of type F_1, F_2 or type F_2, F_1, and cells of the remaining people contained genes of type F_2, F_2. Determine the maximum likelihood estimate of p given n, n_1, n_2. Assume that the type of any one gene in a pair is statistically independent of the other type in the same pair. The gene types across different people are also statistically independent. [10 points]

2. Let Y be a random variable whose values can only be non-negative integers. Assume that $E(Y)$ is well-defined. Then you have to determine which of the following statements is true with full justification. If you feel that none of the statements are true, then state so clearly with full justification. In any case, there will be no points without correct justification. [10 points]

   (a) $E(Y) > \sum_{i=0}^{\infty} P(Y > i)$
   (b) $E(Y) < \sum_{i=0}^{\infty} P(Y > i)$
   (c) $E(Y) = \sum_{i=0}^{\infty} P(Y > i)$

3. Let X_1 and X_2 be two continuous and independent random variables with densities $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ respectively. Derive the CDF and PDF of $X_1 X_2$ (i.e. the product of X_1 and X_2) in terms of $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$. [5+5=10 points]

4. The 'unexpectedness measure' of a discrete random variable X is defined as $U(X) = -\sum_{i=1}^{K} p_i \log p_i$ where $p_i = P(X = i)$ and $\sum_{i=1}^{K} p_i = 1; \forall i, 0 \leq p_i \leq 1$. In this definition, $0 \log 0$ is considered to be 0. For which PMF (i.e. for what values of $\{p_i\}_{i=1}^{K}$) will the unexpectedness measure be maximum? What is this maximum value? Derive your answer by setting the first derivatives of the unexpectedness measure to 0. Obtain the sign of the second derivatives, i.e. sign of $\frac{\partial^2 U}{\partial p_i^2}$. (Note: Given what you have learned so far, you will be able to find only a local maximum. But it turns out that the unexpectedness measure is a concave function, due to which a local maximum is also the global maximum. You are not expected to prove that it is a concave function.) For what PMF, will the unexpectedness measure be the least? Give an intuitive answer (it is not so easy to prove your answer for the minimum, in a quiz/exam). What is this minimum value? [5+1+2+1+1=10 points]

5. Let $Z_1, Z_2, ..., Z_n$ be n independent and identically distributed random variables, belonging to PDF $f_Z(z;\rho)$ having a single parameter ρ. The PDF is defined in the following manner. If $\rho = 0$, then $f_Z(z;\rho) = 1$ if $0 < z < 1$ and 0 otherwise. On the other hand if $\rho = 1$, then $f_Z(z;\rho) = 1/(2\sqrt{z})$ if $0 < z < 1$ and 0 otherwise. The parameter ρ will not acquire any other values. Determine the parameter ρ using maximum likelihood given the values of $Z_1, Z_2, ..., Z_n$. [10 points]